

Smart Christmas Lighting

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Project summary

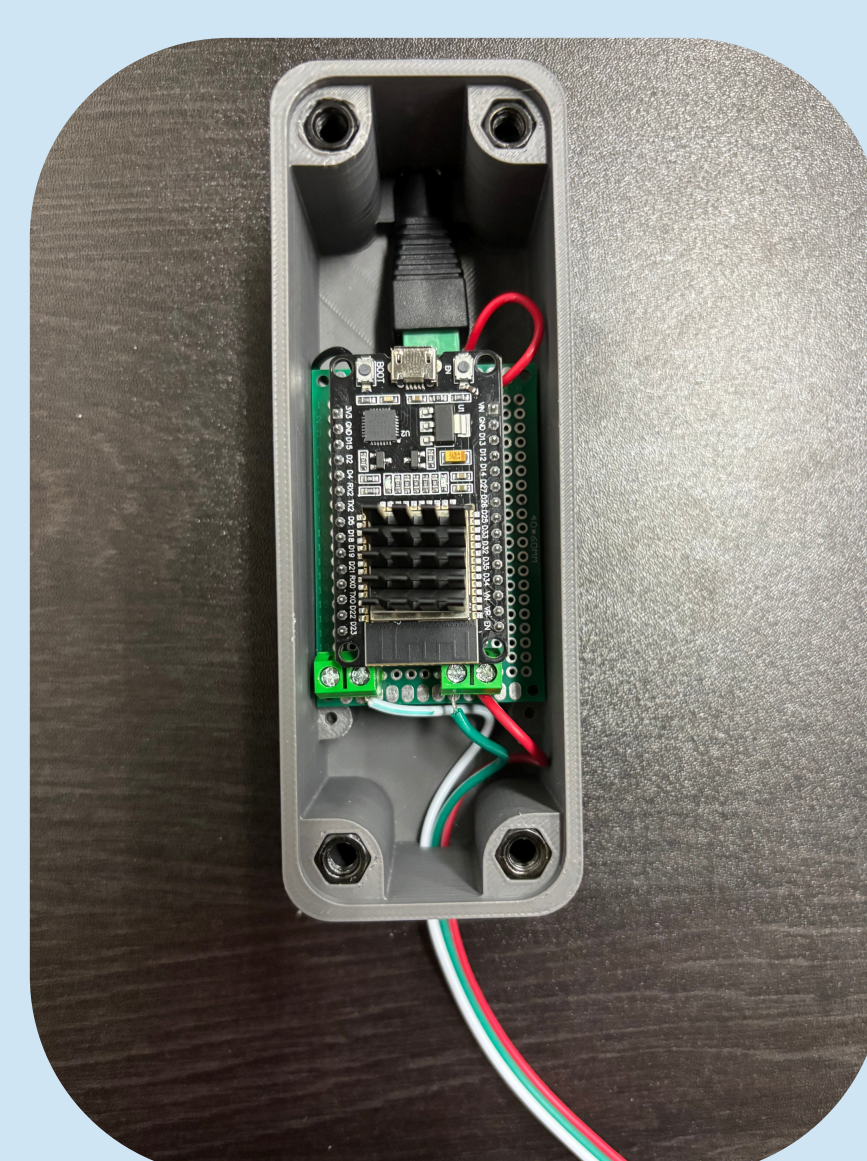
We were tasked with creating a music reactive “smart” Christmas lighting system for downtown Hilliard. Our boxes will be placed at Christmas trees and lights will be connected to them. Each box has a microcontroller that can communicate over WIFI with a central MQTT server to receive lighting instructions. From there the microcontroller processes the information into pulses on the attached lights.



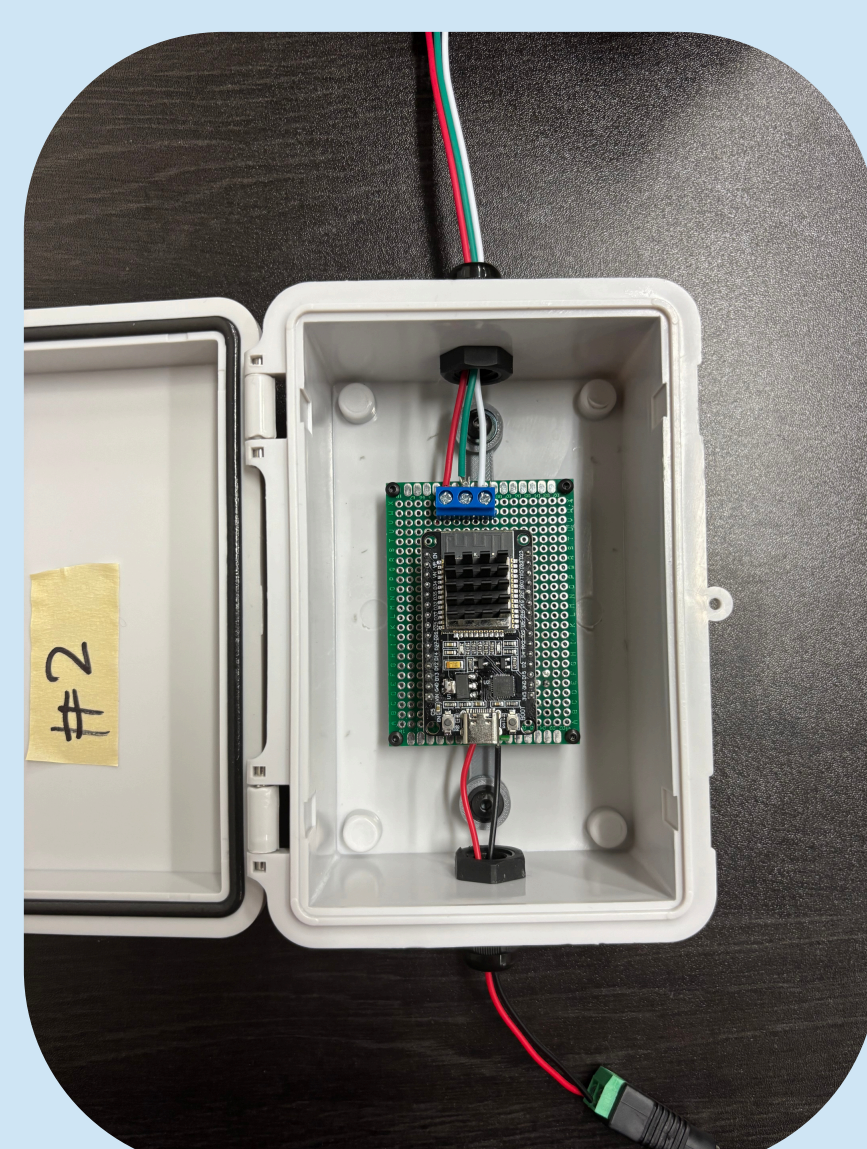
Server Explanation

Our server communicates to the devices through a protocol called MQTT. We chose this protocol because it is very scalable, flexible, relatively quick, and understandable. MQTT has a central broker (server) and clients (users). Each “client” has to subscribe to a topic in order to see the messages sent on that topic, to create a subtopic we can use “/” as in “topic/subtopic”, we use this to individually address boxes in the “lights” topic (“lights/RGB/box1”). MQTT also allows us to add a unique login for each element of our project, allowing us to secure our server

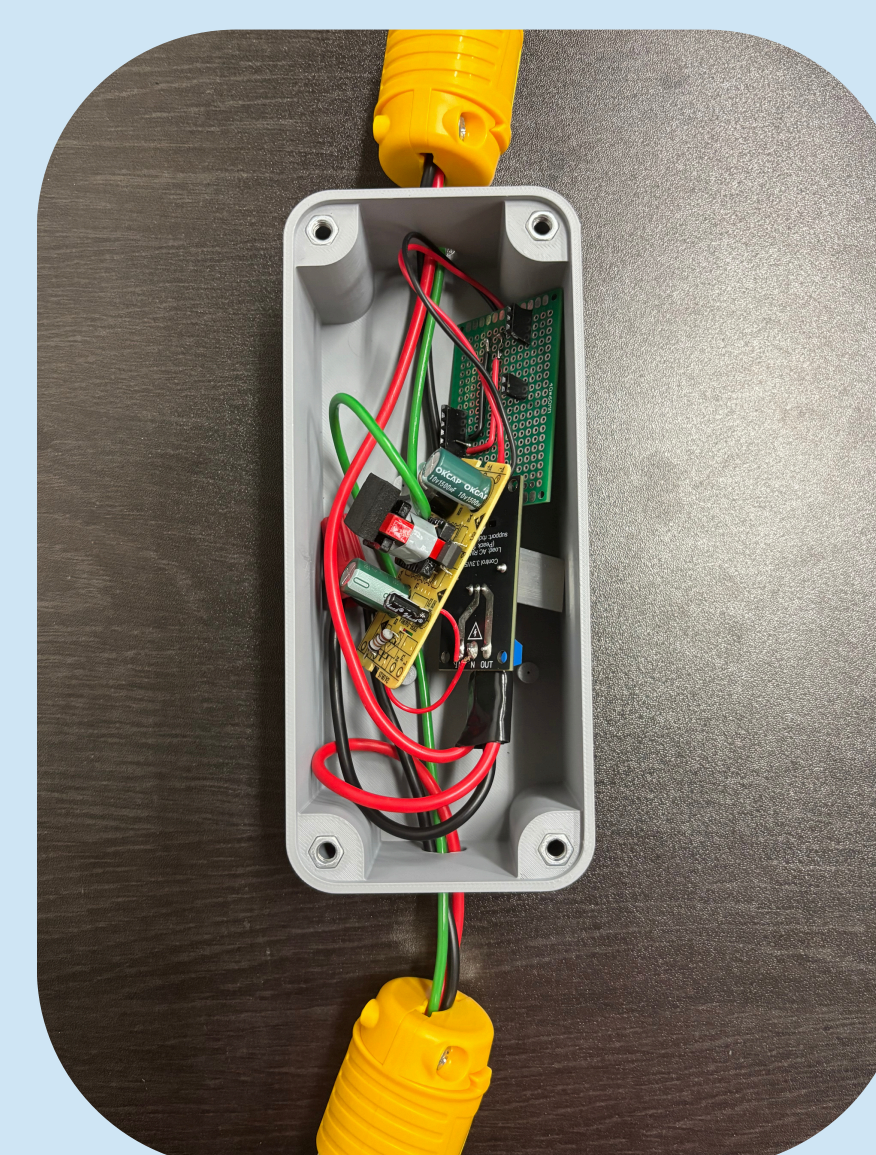
Development Process



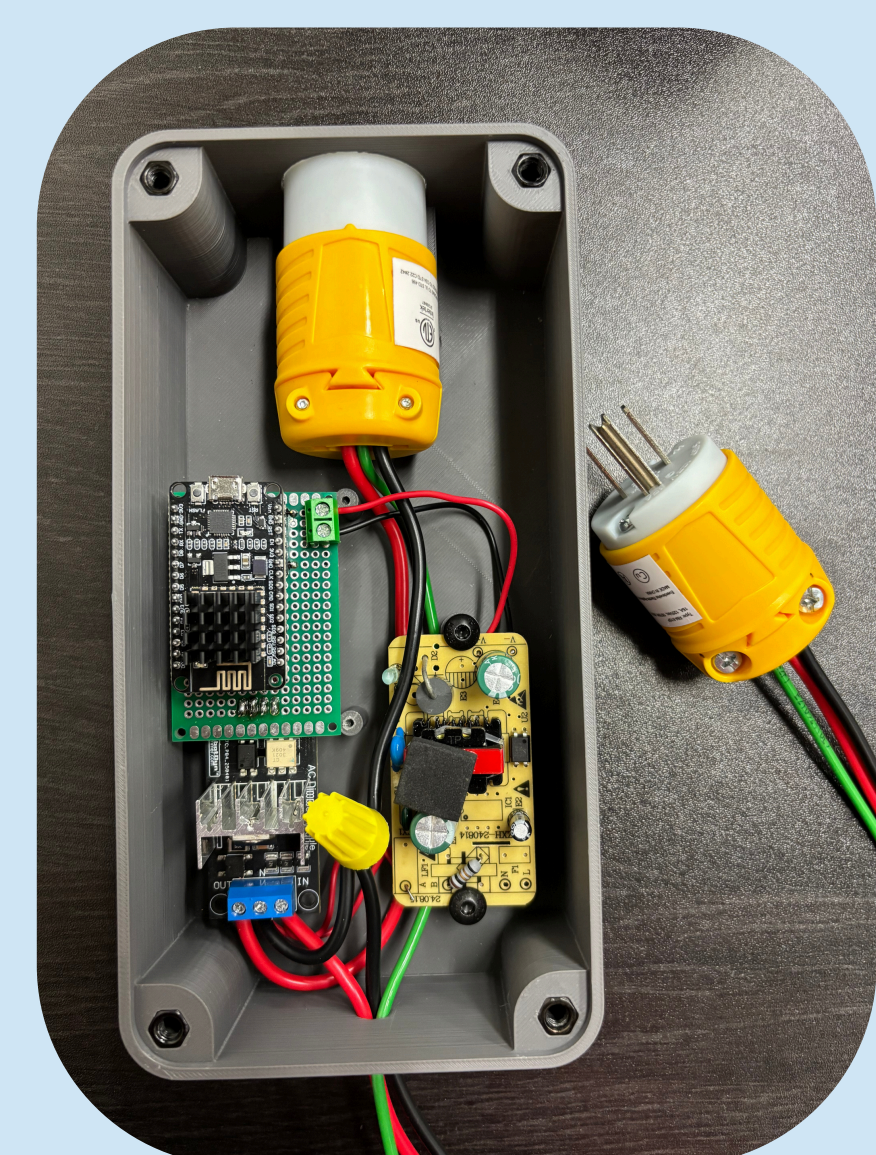
The first RGB box was fully functional. The only issues present were the lack of weatherproofing and the barrel jack input being improperly secured



The second RGB box solves all the old issues but presents new ones. The protoboard cannot handle the power (<3A) needed for 30+ ft of lights, however smaller lengths still work



The first regular lighting box prototype had lacked internal organization and a solid place for the outlets to sit. This prototype was however, fully functional



The second regular lighting box prototype was organized, and had an inset outlet, but could only control one string of lights. The big Hilliard Christmas tree requires six outputs.



The final product is a box using all the same electronics but scaled up to be able to support six outlets on the box. It also uses a more durable extension chord for connection.

System Flow

Audio

A computer plays music and our code detects specific cues to isolate the “beat” of the music, then sends a trigger to the server with each “beat”

Node-RED

Depending on the toggles selected by the user, Node-RED will send different command signals upon receiving the trigger signal from the server

Regular Lighting

An ESP8266 microcontroller receives the signal from Node-RED via Wi-Fi and sends a signal to make the plug in wall lights flash

RGB

An ESP32 microcontroller receives the command signal from Node-RED via Wi-Fi and displays the pulse corresponding to the command signal on the RGB light strip

What's Next?

The last step of our project is to actually implement this system in downtown Hilliard. We will do this some time before Christmas and set up the lighting using the downtown speaker system and Wi-Fi

Acknowledgements

James Desenberg	DJ Daugherty
Joshua DeRee	Jax Stitt
Henk DeRee	Will Sargent
Duane Powell	John Bair
Adam Maynard	Landon Tipword
Mark Tremayne	Dave Jennings
Dave Stewart	Angela Zody

Project Video

